16.1: DNA is the genetic material

Today, even schoolchildren have heard of DNA, and scientists routinely manipulate DNA in the laboratory. Early in the 20th century, however, identifying the molecules of inheritance posed a major challenge to biologists.

The Search for the Genetic Material: *Scientific Inquiry*

Once T. H. Morgan's group showed that genes exist as parts of chromosomes (described in Concept 15.1), the two chemical components of chromosomes—DNA and protein—emerged as the leading candidates for the genetic material. Until the 1940s, the case for proteins seemed stronger: Biochemists had identified proteins as a class of macromolecules with great heterogeneity and specificity of function, essential requirements for the hereditary material. Moreover, little was known about nucleic acids, whose physical and chemical properties seemed far too uniform to account for the multitude of specific inherited traits exhibited by every organism. This view gradually changed as the role of DNA in heredity was worked out in studies of bacteria and the viruses that infect them, systems far simpler than fruit flies or humans. Let's trace the search for the genetic material as a case study in scientific inquiry.

Evidence That DNA Can Transform Bacteria

In 1928, a British medical officer named Frederick Griffith was trying to develop a vaccine against pneumonia. He was studying *Streptococcus pneumoniae*, a bacterium that causes pneumonia in mammals. Griffith had two strains (varieties) of the bacterium, one pathogenic (disease-causing) and one nonpathogenic (harmless). He was surprised to find that when he killed the pathogenic bacteria with heat and then mixed the cell remains with living bacteria of the nonpathogenic strain, some of the living cells became pathogenic **(Figure 16.2)**. Furthermore, this newly acquired trait of pathogenicity was inherited by all the descendants of the transformed bacteria. Apparently, some chemical component of the dead pathogenic cells caused this heritable change, although the identity of the substance was not known. Griffith called the phenomenon ****transformation**, now defined as a change in genotype and phenotype due to the assimilation of external DNA by a cell. Later work by Oswald Avery, Maclyn McCarty, and Colin MacLeod identified the transforming substance as DNA.

Figure 16.2

Inquiry: Can a genetic trait be transferred between different bacterial strains?

Experiment

Frederick Griffith studied two strains of the bacterium *Streptococcus pneumoniae*. The S (smooth) strain can cause pneumonia in mice; it is pathogenic because the cells have an outer capsule that protects them from an animal's immune system. Cells of the R (rough) strain lack a capsule and are nonpathogenic. To test for the trait of pathogenicity, Griffith injected mice with the two strains:

Can a genetic trait be transferred between different bacterial strains?

The experiment has living S cells, living R cells, heat killed S cells, and a mixture of heat killed S cells and living R cells that are all injected into mice.

Living S cells with outer capsule (pathogenic control) injected into mouse. Result: Mouse dies.

Living R cells with no outer capsule (nonpathogenic control) injected into mouse. Result: mouse healthy.

Heat killed S cells in capsule (nonpathogenic control) injected into mouse. Result: Mouse healthy.

Mixture of heat killed S cells in capsule and living R cells without capsule injected into mouse. Result: mouse dies, and living S cells in capsules were found in mouse's blood. These cells could reproduce, producing more S cells.

Results

The result of an experiment on whether a genetic trait can be transferred between different bacterial strains.

The experiment has living S cells, living R cells, heat killed S cells, and a mixture of heat killed S cells and living R cells that are all injected into mice.

Living S cells with outer capsule (pathogenic control) injected into mouse. Result: Mouse dies.

Living R cells with no outer capsule (nonpathogenic control) injected into mouse. Result: mouse healthy.

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Mixture of heat killed S cells in capsule and living R cells without capsule injected into mouse. Result: mouse dies, and living S cells in capsules were found in mouse's blood. These cells could reproduce, producing more S cells.

Conclusion

The living R bacteria had been transformed into pathogenic S bacteria by an unknown, heritable substance from the dead S cells that enabled the R cells to make capsules.

Data from F. Griffith, The significance of pneumococcal types, *Journal of Hygiene* 27:113–159 (1928).

WHAT IF? *How did this experiment rule out the possibility that the R cells simply used the dead S cells' capsules to become pathogenic?*

*For suggested answer, see*Appendix A*.*

Scientists remained skeptical, however, since many still viewed proteins as better candidates for the genetic material. Also, many biologists were not convinced that bacterial genes would be similar in composition and function to those of more complex organisms. But the major reason for the continued doubt was that so little was known about DNA.

Evidence That Viral DNA Can Program Cells

Additional evidence that DNA was the genetic material came from studies of viruses that infect bacteria **(Figure 16.3)**. These viruses are called **bacteriophages** (meaning "bacteria-eaters"), or **phages** for short. Viruses are much simpler than cells. A **virus** is little more than DNA (or sometimes RNA) enclosed by a protective coat, which is often simply protein. To produce more viruses, a virus must infect a cell and take over the cell's metabolic machinery.

Figure 16.3

A virus infecting a bacterial cell.

Phage head with D N A inside, above tail sheath and tail fibers, injecting genetic material into a bacterial cell. Scale: 100 nanometers.

A phage called T2 attaches to a host cell and injects its genetic material through the plasma membrane, while the head and tail parts remain on the outer bacterial surface (colorized TEM).

Phages have been widely used as tools by researchers in molecular genetics. In 1952, Alfred Hershey and Martha Chase performed experiments showing that DNA is the genetic material of a phage known as T2. This is one of many phages that infect *Escherichia coli* (*E. coli*), a bacterium that normally lives in the intestines of mammals and is a model organism for molecular biologists. At that time, biologists already knew that T2, like many other phages, was composed almost entirely of DNA and protein. They also knew that the T2 phage could quickly turn an *E. coli* cell into a T2-producing factory that released many copies of

new phages when the cell ruptured. Somehow, T2 could reprogram its host cell to produce viruses. But which viral component—protein or DNA—was responsible?

Hershey and Chase answered this question by devising an experiment showing that only one of the two components of T2 actually enters the *E. coli* cell during infection **(Figure 16.4)**. In their experiment, they used a radioactive isotope of sulfur to tag protein in one batch of T2 and a radioactive isotope of phosphorus to tag DNA in a second batch. Because protein, but not DNA, contains sulfur, radioactive sulfur atoms were incorporated only into the protein of the phage. In a similar way, the atoms of radioactive phosphorus labeled only the DNA, not the protein, because nearly all the phage's phosphorus is in its DNA. In the experiment, separate samples of nonradioactive *E. coli* cells were infected separately with the protein-labeled and DNA-labeled batches of T2. The researchers then tested the two samples shortly after the onset of infection to see which type of radioactively labeled molecule—protein or DNA—had entered the bacterial cells and would therefore be capable of reprogramming them.

Figure 16.4

Inquiry: Is protein or DNA the genetic material of phage T2?

Experiment

Alfred Hershey and Martha Chase used radioactive sulfur and phosphorus to trace the fates of protein and DNA, respectively, of T2 phages that infected bacterial cells. They wanted to see which of these molecules entered the cells and could reprogram them to make more phages.

Experiment. Is protein or D N A the genetic material of phage T 2?

Batch 1: Phages were grown with radioactive sulfur (superscript 35 S), which was incorporated into the phage protein (pink fluid inside a flask).

- 1. Inside flask, researchers mixed radioactively labeled phages with bacteria. The phages infected the bacterial cells. Bacterial cell drawn with phages (containing D N A) and radioactive proteins on the surface.

- 2. The mixture from the flask was agitated in a blender to free phage parts outside the bacteria from the cells. Drawing shows bacterial cell with phage D N A inside and empty protein shells and phages outside cell coming loose from bacterial cell.

- 3. Mixture centrifuged so bacteria (with phage D N A) formed a pellet at bottom of a test tube; free phages and page parts, which are lighter, remained suspended in the liquid.
- 4. Radioactivity measured in pellet and liquid; found in liquid (from phage protein).

Batch 2: Phages were grown with radioactive phosphorous (superscript 32 P), which was incorporated into phage D N A (blue parts in phages drawn and in liquid in a flask). Phages containing radioactive D N A (blue) shown in drawing on a bacterial cell.

- Mixture from flask agitated in blender, causing phages to come lose from outside bacterial cells, while bacterial cells contain phage D N A inside.
- Mixture from blender centrifuged; pellet at the bottom of a test tube contains bacteria and liquid contains free phages and phage parts, which are lighter.
- Radioactivity (phage D N A) found in pellet (because phage D N A is within bacterial cells).

Results. When proteins were labeled (batch 1, radioactivity remained outside the cells, but when D N A was labeled (batch 2, radioactivity was found inside the cells. Cells containing radioactive phage D N A released new phages with some radioactive phosphorus. Conclusion. Phage D N A entered bacterial cells, but phage proteins did not. Hershey and Chase concluded that D N A, not protein, functions as the genetic material of phage T 2.

Results

When proteins were labeled (batch 1), radioactivity remained outside the cells, but when DNA was labeled (batch 2), radioactivity was found inside the cells. Cells containing radioactive phage DNA released new phages with some radioactive phosphorus.

Conclusion

Phage DNA entered bacterial cells, but phage proteins did not. Hershey and Chase concluded that DNA, not protein, functions as the genetic material of phage T2.

Data from A. D. Hershey and M. Chase, Independent functions of viral protein and nucleic acid in growth of bacteriophage, *Journal of General Physiology* 36:39–56 (1952).

WHAT IF? *How would the results have differed if proteins carried the genetic information?*

*For suggested answer, see*Appendix A*.*

Animation

The Hershey-Chase Experiment

image

Play

image

Hershey and Chase found that the phage DNA entered the host cells, but the phage protein did not. Moreover, when these bacteria were returned to a culture medium and the infection ran its course, the *E. coli* released phages that contained some radioactive phosphorus. This result further showed that the DNA inside the cell played an ongoing role during the infection process. They concluded that the DNA injected by the phage must be the molecule carrying the genetic information that makes the cells produce new viral DNA and proteins. The Hershey-Chase experiment was a landmark study because it provided powerful evidence that nucleic acids, rather than proteins, are the hereditary material, at least for certain viruses.

Additional Evidence That DNA Is the Genetic Material

Further evidence that DNA is the genetic material came from the laboratory of biochemist Erwin Chargaff. DNA was known to be a polymer of nucleotides, each having three components: a nitrogenous (nitrogen-containing) base, a pentose sugar called deoxyribose, and a phosphate group **(Figure 16.5)**. The base can be adenine (A), thymine (T), guanine (G), or cytosine (C). Chargaff analyzed the base composition of DNA from a number of different organisms. In 1950, he reported that the base composition of DNA varies from one species to another. For example, he found that 32.8% of sea urchin DNA nucleotides have the base A, whereas only 24.7% of those from the bacterium *E. coli* have an A. This evidence of molecular diversity among species, which most scientists had presumed to be absent from DNA, made DNA a more credible candidate for the genetic material.

Figure 16.5

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The structure of a DNA strand.
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D N A strand from 5 prime end to 3 prime. From 5 prime end: backbone alternates phosphate group -sugar-phosphate group-sugar; a base (A, T, C, G) is attached to each sugar.

D N A consists of a phosphate-sugar backbone with bases (thymine, guanine, cytosine, or adenine) bonded to the sugars. Each repeating nucleotide monomer is made up of one phosphate group, one sugar, and one base. A phosphate group is a central P double bonded to one O and single bonded to 3 other Os. Two Os have

negative charges. The third is bonded to the fifth C in deoxyribose (pentose sugar). Attached to carbon 1 of the sugar is the base (A, T, C or G). Attached to carbon 3 of the sugar is the next phosphate group in the backbone of the molecule. This structure repeats for the rest of the chain. The strand of D N A shown starts at the 5 prime end (with the first phosphate group) and ends at the 3 prime end (with the O H group of the last sugar). The base sequence shown here is T (single ring base), G (double ring base), C (single ring base), and A (double ring base).

Each DNA nucleotide monomer consists of the sugar deoxyribose (blue) attached to both a nitrogenous base (A, T, G, or C), and a phosphate group (yellow). The phosphate group of one nucleotide is attached to the sugar of the next by a covalent bond, forming a “backbone” of alternating phosphates and sugars from which the bases project. A polynucleotide strand has directionality, from the 5′ end (with the phosphate group) to the 3′ end (with the —OH group of the sugar). 5′ and 3′ refer to the numbers assigned to the carbons in the sugar ring (see magenta numbers).

Animation

DNA and RNA Structure

image

Play

image

Chargaff also noticed a peculiar regularity in the ratios of nucleotide bases. In the DNA of each species he studied, the number of adenines approximately equaled the number of thymines, and the number of guanines approximately equaled the number of cytosines. In sea urchin DNA, for example, the four bases are present in these percentages: A=32.8% and T=32.1%; G=17.7% and C=17.3%. (The percentages are not exactly the same because of limitations in Chargaff’s techniques.)

These two findings became known as *Chargaff’s rules*: (1) DNA base composition varies between species, and (2) for each species, the percentages of A and T bases are roughly equal, as are those of G and C bases. In the **Scientific Skills Exercise**, you can use Chargaff’s rules to predict percentages of nucleotide bases. The basis for these rules remained unexplained until the discovery of the double helix.

Scientific Skills Exercise

Working with Data in a Table

AP Science Practices 1, 2, 4, 5, 6

Given the Percentage Composition of One Nucleotide in a Genome, Can We Predict the Percentages of the Other Three Nucleotides? Even before the structure of DNA was elucidated, Erwin Chargaff and his co-workers noticed a pattern in the base composition of nucleotides from different species: The percentage of adenine (A) bases roughly equaled that of thymine (T) bases, and the percentage of cytosine (C) bases roughly equaled that of guanine (G) bases. Further, the percentage of each pair (A–T or C–G) varied from species to species. We now know that the 1:1 A/T and C/G ratios are due to complementary base pairing between A and T and between C and G in the DNA double helix, and differences between species are due to the unique sequences of bases along a DNA strand. In this exercise, you will apply Chargaff's rules to predict the composition of bases in a genome.

How the Experiments Were Done In Chargaff's experiments, DNA was extracted from the given species, hydrolyzed to break apart the nucleotides, and then analyzed chemically. These studies provided approximate values for each type of nucleotide. (Today, whole-genome sequencing allows base composition analysis to be done more precisely directly from the sequence data.)

Data from the Experiments Tables are useful for organizing sets of data representing a common set of values (here, percentages of A, G, C, and T) for a number of different samples (in this case, from different species). You can apply the patterns that you see in the known data to predict unknown values. In the table, complete base distribution data are given for sea urchin DNA and salmon DNA; you will use Chargaff's rules to fill in the rest of the table with predicted values.

Sea urchin.

Sea urchin

Predicting composition of nucleotides in genomes with the composition of one nucleotide.

Data from several papers by Chargaff: for example, E. Chargaff et al., Composition of the desoxypentose nucleic acids of four genera of sea-urchin, *Journal of Biological Chemistry* 195: 155–160 (1952).

Source of D N A	Base percentage
Adenine	Guanine
Sea urchin	32.8
Salmon	29.7
Wheat	28.1

E. coli	24.7
Human	30.4
Ox	29
Average percentage	blank

INTERPRET THE DATA

- Explain how the sea urchin and salmon data demonstrate both of Chargaff's rules.
- Using Chargaff's rules, fill in the table with your predictions of the missing percentages of bases, starting with the wheat genome and proceeding through *E. coli*, human, and ox. Show how you arrived at your answers.
- If Chargaff's rule—that the amount of A equals the amount of T and the amount of C equals the amount of G—is valid, then hypothetically we could extrapolate this to the combined DNA of all species on Earth (like one huge Earth genome). To see whether the data in the table support this hypothesis, calculate the average percentage for each base in your completed table by averaging the values in each column. Does Chargaff's equivalence rule still hold true?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

HHMI BioInteractive Animation

Chargaff's Ratio

image

Play

image

For more resources from HHMI BioInteractive, go to www.biointeractive.org.

Building a Structural Model of DNA

Once most biologists were convinced that DNA was the genetic material, the challenge was to determine how the structure of DNA could account for its role in inheritance. By the early 1950s, the arrangement of covalent bonds in a nucleic acid polymer was well established (see Figure 16.5), and researchers focused on discovering the three-dimensional structure of DNA. Among the scientists working on the problem were Linus Pauling, at the California Institute of Technology, and Maurice Wilkins and Rosalind Franklin, at

King's College in London. First to come up with the complete answer, however, were two scientists who were relatively unknown at the time—the American James Watson and the Englishman Francis Crick.

The collaborative work that solved the puzzle of DNA structure began soon after Watson went to Cambridge University, where Crick was studying protein structure with a technique called X-ray crystallography (see Figure 5.21). While visiting the laboratory of Maurice Wilkins, Watson saw an X-ray diffraction image of DNA produced by Wilkins's accomplished colleague Rosalind Franklin **(Figure 16.6)**. Images produced by X-ray crystallography are not actually pictures of molecules. The spots in the image were produced by X-rays that were diffracted (deflected) as they passed through aligned fibers of purified DNA. Watson was familiar with the type of X-ray diffraction pattern of helical molecules, and seeing the photo that Wilkins showed him confirmed that DNA was helical in shape. The photo also added to earlier data obtained by Franklin and others suggesting the width of the helix and the spacing of the nitrogenous bases along it. The pattern in this photo implied that the helix was made up of two strands, contrary to a three-stranded model proposed by Linus Pauling a short time earlier. The presence of two strands accounts for the now-familiar term ****__double helix__. DNA is shown in some of its many different representations in **Figure 16.7**.

Figure 16.6

Rosalind Franklin and her X-ray diffraction photo of DNA.

Figure a is Rosalind Franklin. Figure b is Franklin's X-ray diffraction of D N A. It has spaced segments in an X shape.

Advanced placement edition

Apply concepts and connect across Big Ideas using visual representations.

How does this model support common ancestry for all organisms? Given a distance of 0.34 nm between nucleotides, how long is the haploid DNA of a human with 3.2 billion nucleotides?

Figure 16.7

Visualizing DNA

D N A can be illustrated in many ways, but all diagrams represent the same basic structure. This figure compares structural images, simplified images, and D N A sequences.

These structural images show the three-dimensional shape of the D N A double helix (left) and chemical details of D N A's structure (right). Both images use the same colors for phosphate groups (yellow), deoxyribose sugars (blue), and nitrogenous bases (shades of green and orange).

Structural image on left: The D N A double helix is right-handed, as shown in this computer-generated space-filling model. Use your right hand (as shown in photo) to grasp the model and follow the sugar-phosphate backbone up the helix (red arrow) and around to the back. (It won't work with your left hand.) Phosphate groups are yellow, sugars are blue, and bases are dark orange, light orange, dark green and light green. Components tightly packed and twisted into double helix. Bases are 0.34 nanometers apart. There is one full turn every 10 base pairs (3.4 nanometers).

Structural image on right: Here, the two D N A strands are shown untwisted so it’s easier to see the chemical details. Note that the strands are antiparallel, they are oriented in opposite directions, like the lanes of a divided street. Sugar-phosphate backbone on left runs (from top to bottom) 5 prime to three prime; backbone on the right runs from (top to bottom) 3 prime to 5 prime. Strands consist of repeating nucleotides (phosphate-sugar-base) with covalent bonds (solid lines) drawn between the phosphate group, sugar and base and hydrogen bonds (dashed lines) drawn between complementary bases (which form the rungs of the helix ladder.)

In this diagram, from the top left, is a phosphate group (5 prime end), pentose sugar, and T (single ring) hydrogen bonded (two dashed lines) to A (double ring), which is bonded to a pentose sugar (3 prime end), then a sugar-phosphate backbone.

Back to the left side: phosphate, sugar, G (double ring) hydrogen bonded (3 dashed lines) to a C (single ring), which is bonded to a sugar and then a phosphate group. Covalent sugar-phosphate bonds link the nucleotides of each strand.

Back to the left side: phosphate group-sugar-C (single ring) hydrogen bonded (3 dashed lines) to a G, then sugar-phosphate group. Hydrogen bonds (shown as dotted lines) between nitrogenous bases hold the strands together.

Back to the left side: phosphate group- sugar (3 prime end) bonded to an A (double ring) hydrogen bonded (2 dashed lines) to a T, then a sugar-phosphate group (5 prime end). Van der Waals interactions between stacked base pairs help hold the molecule together. When molecular detail is not necessary, D N A is portrayed in a range of simplified diagrams, depending on the focus of the figure. From left to right: ribbon models, ladder-style diagrams, and straight lines.

Ribbon models. The ribbons in these simplified double helix diagrams represent the sugar-phosphate backbones. Sugar-phosphate backbone shown as double strand of blue ribbon with bases on the inside. H-bonds drawn as dashed lined between complementary bases (A with T; and C with G).

Ladder-style diagrams. These flattened ladder style diagrams of D N A depict the sugar-phosphate backbones like the side rails of a ladder, with the base pairs as rungs. Light blue is used to indicate the more

recently synthesized strand. Shown here are three ladder-style diagrams. In each diagram, each horizontal rung contains two base pairs: T with A, G with C, C with G, and A with T.

Straight lines. Sometimes the double-stranded D N A molecule is shown simply as two straight lines. Diagram shows two thin parallel vertical lines. The line on the left is labeled 5 prime at the top and 3 prime at the bottom. The line on the right is labeled 3 prime at the top and 5 prime at the bottom.

Genetic information is carried in D N A as a linear sequence of nucleotides that may be transcribed into m R N A and translated into a polypeptide. When focusing on the D N A sequence, each nucleotide can be represented simply as the letter of its base: A, T, C, or G. In this example, starting from the left: 3 prime, A C G T A A G C G G T T A A T, 5 prime. On the next line, starting from the left: 5 prime, T G C A T T C G C C A A T T A, 3 prime.

Animation

DNA Double Helix

image

Play

image

Instructors: Additional questions related to this Visualizing Figure can be assigned in **Mastering Biology**.

Watson and Crick began building models of a double helix that would conform to the X-ray measurements and what was then known about the chemistry of DNA, including Chargaff's rule of base equivalences. Having also read an unpublished annual report summarizing Franklin's work, they knew she had concluded that the sugar-phosphate backbones were on the outside of the DNA molecule, contrary to their working model. Franklin's arrangement was appealing because it put the negatively charged phosphate groups facing the aqueous surroundings, while the relatively hydrophobic nitrogenous bases were hidden in the interior. Watson constructed such a model, in which the two sugar-phosphate backbones are ****antiparallel—that is, their subunits run in opposite directions (see Figure 16.7). You can imagine the overall arrangement as a rope ladder with rigid rungs. The side ropes represent the sugar-phosphate backbones, and the rungs represent pairs of nitrogenous bases. Now imagine twisting the ladder to form a helix. Franklin's X-ray data indicated that the helix makes one full turn every 3.4 nm along its length. With the bases stacked just 0.34 nm apart, there are ten layers of base pairs, or rungs of the ladder, in each full turn of the helix.

The nitrogenous bases of the double helix are paired in specific combinations: adenine (A) with thymine (T), and guanine (G) with cytosine (C). It was mainly by trial and error that Watson and Crick arrived at this key feature of DNA. At first, Watson imagined that the bases paired like with like—for example, A with A and C

with C. But this model did not fit the X-ray data, which suggested that the double helix had a uniform diameter. Why is this requirement inconsistent with like-with-like pairing of bases? Adenine and guanine are purines, nitrogenous bases with two organic rings, while cytosine and thymine are nitrogenous bases called pyrimidines, which have a single ring. Pairing a purine with a pyrimidine is the only combination that results in a uniform diameter for the double helix **(Figure 16.8)**.

Figure 16.8

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Possible base pairings in the DNA double helix.
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Dashed red lines mark width of D N A from X-ray data. Purine + pyrimidine (double ring + single ring) is the only combination of bases that is consistent with X-ray data.

Dashed red lines show the width of D N A from X-ray data. Purine + purine = double ring + double ring = too wide (wider than dashed lines). Pyrimidine + pyrimidine = single ring + single ring = too narrow (narrower than dashed lines). Purine + pyrimidine = fits dashed lines and is consistent with X-ray data.

Watson and Crick reasoned that there must be additional specificity of pairing dictated by the structure of the bases. Each base has chemical side groups that can form hydrogen bonds with its appropriate partner: Adenine can form two hydrogen bonds with thymine and only thymine; guanine forms three hydrogen bonds with cytosine and only cytosine. In shorthand, A pairs with T, and G pairs with C **(Figure 16.9)**.

Figure 16.9

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Base pairing in DNA.
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Two black dotted lines (hydrogen bonds) form between Adenine and Thymine, and three black dotted lines (hydrogen bonds) form between Guanine and Cytosine.

Hydrogen bonds form between chemical side groups of complementary nitrogenous base pairs in the D N A molecule. Between A and T, hydrogen bonds form between an H on A and an O on T, and between an N on A and an H on T. Two total hydrogen bonds form between A and T. Between G and C, hydrogen bonds from between an O on G and an H on C, and between an H on G and an N on C, and between an H on G and an O on C. Three total hydrogen bonds form between G and C.

The Watson-Crick model took into account Chargaff's ratios and ultimately explained them. Wherever one strand of a DNA molecule has an A, the partner strand has a T. Similarly, a G in one strand is always paired with a C in the complementary strand. Therefore, in the DNA of any organism, the amount of adenine equals

the amount of thymine, and the amount of guanine equals the amount of cytosine. (Modern DNA-sequencing techniques have confirmed that the amounts are exactly equal.) Although the base-pairing rules dictate the combinations of nitrogenous bases that form the “rungs” of the double helix, they do not restrict the sequence of nucleotides *along* each DNA strand. The linear sequence of the four bases can be varied in countless ways, and each gene has a unique base sequence.

In April 1953, Watson and Crick surprised the scientific world with a succinct, one-page paper that reported their molecular model for DNA: the double helix, which has since become an icon of molecular biology. Watson and Crick, along with Maurice Wilkins, were awarded the Nobel Prize in 1962 for this work. (Sadly, Rosalind Franklin had died at the age of 37 in 1958 and was thus ineligible for the prize.) The beauty of the double helix model was that the structure of DNA suggested the basic mechanism of its replication.

HHMI BioInteractive Video

Great Discoveries in Science: The Double Helix

image

image

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